

ALEX SANTORA

☎ 617-285-6219 ✉ asantora1@stevens.edu [in linkedin.com/in/alex-santora](https://www.linkedin.com/in/alex-santora) 🏠 Hoboken, NJ

Education

Stevens Institute Of Technology

Bachelor of Science in Computer Engineering

Minor in Pure and Applied Mathematics

Sep. 2025 – May 2029

Hoboken, NJ

Technical Skills

Languages: Python, Java, C#, C++

Tools / Technologies: VS Code, Excel, Google Sheets, Raylib, Unity, Arduino, SolidWorks

Honors and Activities: Dean's List, Edwin A. Scholarship, Alpha Kappa Psi

GPA: 3.88/4.0

Relevant Coursework: Entrepreneurial Thinking, Differential Equations, Linear Algebra, Mechanics, Vectors and Matrices, Multi-variable Calculus

Projects

Autonomous Robot | *Academic*

January 2026

- Designed and developed an autonomous robot in a 6-week team engineering project using a WeMos ESP8266, 3 ultrasonic sensors, 2 DC motors, and an OLED display, combining embedded programming, sensing, and hardware integration to create a fully functional navigation system.
- Implemented Arduino/C++ navigation logic and obstacle-avoidance behavior to drive the robot to 4 designated targets in sequence, using live MQTT-based x-y coordinate feedback to make real-time movement and positioning adjustments during operation.
- Applied SolidWorks design, subsystem integration, debugging, and repeated performance testing to calibrate sensors, tune motion behavior, and improve reliability throughout road tests, speed challenges, prototype reviews, and final project evaluation.

Smart IoT Watering System | *Academic*

September 2025

- Programmed an ESP32 microcontroller in C++/Arduino to collect temperature, humidity, soil moisture, and light data, perform edge-level Fahrenheit conversion, and publish readings to an MQTT cloud server every 10 minutes.
- Designed and 3D-modeled a compact enclosure in SolidWorks to mount sensors, mini pump, and wiring; optimized for material use and print time on Flashforge 3D printers.
- Integrated a moisture-triggered pump control system that dispensed water automatically based on calibrated soil humidity thresholds, then deployed and validated performance with multi-day sensor data analysis in Excel.

Baseball Framing Metrics | *Personal*

November 2025

- Built a catcher framing analytics system in Python, OpenCV, and Excel to review video and chart 1000+ called pitches across 20 games, tracking location, count, result, and umpire call.
- Organized and analyzed data in Excel, creating structured logs for pitch location, count, result, and call outcome to support consistent comparison across games.
- Built a data-driven catcher performance database that established average and replacement-level benchmarks, enabling coaches to evaluate players against standardized metrics and make more informed personnel decisions.

Work Experience

Playland Arcade

(Seasonal) May 2023 – Present

Arcade Attendant

Hampton Beach, NH

- Collaborate with a team to maintain a customer-friendly atmosphere, handling misunderstandings with sensitivity and professionalism to ensure guests leave happy, serving up to 250 guests per hour.
- Manage and handle up to \$5,000 in cash daily, ensuring accuracy and security in all transactions.
- Deliver technical assistance for over 100 arcade games and ticket dispensers, ensuring smooth operation and minimal downtime.

Leadership / Extracurricular

Collegiate NCAA Varsity Baseball Player

September 2025 – Present

Team Member

Stevens Institute of Technology

- Compete as a member of the Stevens Institute of Technology baseball team, balancing the demands of collegiate athletics with a rigorous engineering course load.
- Contribute to a team-first culture by communicating effectively, responding to coaching, and maintaining composure under pressure during practices and competition.